Yuxiang Tian

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Research interests: robot-head mechanisms and control; visual self-modeling; adaptive robot learning.

EDUCATION

Columbia University New York, NY
MS in Mechanical Engineering, GPA: 4.247 Expected June 2027
Selected coursework: Graduate Design; Kinematics of Machines and Robots; Advanced Machine Dynamics; Human-Centered Design and Innovation.

China Three Gorges University Yichang, Hubei, China
B.Eng. in Mechanical Design, Manufacturing and Automation, WES GPA: 3.35/4.0 2024
Selected coursework: Robot Technology and Application; Fluid Mechanics; Higher Mathematics; College Physics.

PUBLICATIONS AND MANUSCRIPTS

Learning Recoverability-Aware, Response-Conditioned Probes for Robot Self-Diagnosis
Yuxiang Tian, Yucheng Yang, Ding Ma
ICRA 2027. Under review. Keywords: active self-diagnosis; response-conditioned probing; recoverability.

Active Self-Diagnosis and Diagnosis-Gated Heterogeneous Recovery for Quadruped Locomotion under Actuator Faults
Yuxiang Tian (sole author)
Engineering Applications of Artificial Intelligence. Under review after revision. Keywords: actuator fault diagnosis; diagnosis-gated recovery; residual reinforcement learning.

Physical probing can shift the target of robot supervision
Yuxiang Tian (sole author)
Communications Engineering. Under review. Keywords: interactive perception; physical probing; robot supervision.

Counterfactual Auditing of Body-Indexed Command Authority in Quadruped Robots
Yuxiang Tian (sole author)
ICRA 2027. Under review. Keywords: body-local control; counterfactual auditing; quadruped control.

Beyond the Latest Checkpoint: Policy-Lifetime Assurance for Continually Adapted Robots
Yuxiang Tian (sole author)
IEEE Transactions on Robotics. Under review. Keywords: continual robot learning; policy auditing.

Staged Cross-View Memory for Pose-Agnostic Robotic Inspection
Yuxiang Tian, Yucheng Yang, Ding Ma
ICRA 2027. Under review. Keywords: robot-camera inspection; multi-view anomaly detection; DINOv3.

Reciprocal Audits Separate Robot Endpoint Classification from Command–Execution Correspondence
Yuxiang Tian, Yucheng Yang, Ding Ma
ICRA 2027. Under review. Keywords: robot task verification; command-execution correspondence.

Frozen V-JEPA 2.1 for Offline Tactile Slip Perception: Label-Efficient Ranking and the Alarm Gap
Yuxiang Tian, Yucheng Yang, Ding Ma
ICRA 2027. Under review. Keywords: tactile slip perception; label-efficient transfer; V-JEPA; alarm calibration.

Locally Optimal Aerodynamic Shape Design for Stratospheric Airships: A GA-Driven MDO Framework with CAD/CFD Validation
Yuxiang Tian (sole author)
Aerospace Science and Technology 168, 110965 (2026). DOI: 10.1016/j.ast.2025.110965. Keywords: GA-MDO; CAD/CFD validation.

Robot Skill Compiler: An Auditable Workbench for Language-Specified MuJoCo Environments
Yuxiang Tian (sole author)
Accepted, IROS 2026 Workshop on Building Scalable Infrastructure for Robot Learning: From Data Scaling to Real-World Deployment. Keywords: MuJoCo; task compilation; reproducibility.

Structural Design and Optimization of Egg Carrier for Dynamic Egg Slit Detection Platforms
Ronghua Meng, **Yuxiang Tian** (joint senior and co-corresponding author), Siwei Huang
PLOS ONE 20(4), e0320848 (2025). DOI: 10.1371/journal.pone.0320848. Keywords: GA optimization; structural design; FEA/fatigue.

Disassembly Sequence Optimization of Lithium Battery Pack Based on Priority Constraint Graph
Ronghua Meng, **Yuxiang Tian**, Ningjing Shu, Liuyang Cai, Huawei Song, Yuping Zhang
Journal of Physics: Conference Series 2650, 012021 (2023). DOI: 10.1088/1742-6596/2650/1/012021. Keywords: priority constraint graph; SA-PSO; battery disassembly planning.

Calibration-Free Analyte Measurements Using Aptameric Graphene Nanosensors
Wenting Dai, Yuchen Yang, **Yuxiang Tian**, Kyung-Ae Yang, Milan N. Stojanovic, Qiao Lin
Manuscript in preparation. Keywords: aptameric nanosensors; kinetic modeling; vancomycin assay.

RESEARCH AND TEACHING EXPERIENCE

Columbia Creative Machines Laboratory | Research Assistant **Mar 2026 - Present**
Advisor: Professor Hod Lipson

- Redesigned an **81-servo robotic head** for coordinated facial expressions, including actuator layout, Bowden-cable routing and elastic returns in the head and chest; refined mechanisms for lip pressing and puckering.
- Completed detailed CAD and preliminary mechanism and facial-skin simulations; refining the design for fabrication, calibration and expression-control testing.
- Developing xArm6 fault diagnosis in simulation using wrist-camera RGB, optical flow and position-specific healthy references, without motor signals. Generated 47,520 trajectories; exploratory development evaluation achieved 88.84% end-to-end fault localization accuracy and 87.73% accuracy for identifying both fault type and affected joint.

Robot Self-Diagnosis and Recovery | Research Lead

- Proposed and led response-conditioned diagnostic probing across xArm6, xArm5 and Go2 simulations. In a matched two-step xArm6 evaluation, identified both fault type and affected joint with 83.23% accuracy, versus 77.01% after shuffling response-action correspondence.
- Developed diagnosis-gated quadruped recovery using delay prediction, analytical compensation and residual reinforcement learning. With known fault descriptors, increased success over 10 m terrain under delay faults from 49.2% to 80.2% in simulation.

Columbia BioMEMS Laboratory | Research Assistant **Mar 2026 - Present**
Advisor: Professor Qiao Lin

- Developing closed-loop control for fluidic dispensing and supporting small-molecule detection using electrical readouts from aptameric nanosensors.
- Conducting molecular dynamics simulations of aptamer-target interactions to complement wet-lab experiments; provided research guidance to two undergraduate students.

Columbia University | Teaching Assistant, MECE4213E **Fall 2026**

- Graded assignments, answered students' questions and provided course-project feedback for Lab-on-a-Chip and Microrobotic Devices.

Columbia Makerspace | Student Casual – Admin

- Led AI-assisted design and integration of a Jetson Orin Nano dual-camera occupancy prototype using DeepStream, PeopleNet and TensorRT, with live zone counts and 15-minute CSV logging. Corrected camera encoding settings, improving preview from ~6 to 17–19 FPS (detection updates: 2–3 FPS); verified functionality on hardware.

ADDITIONAL INFORMATION

Academic service: Peer Reviewer, Advances in Space Research (Aug - Sep 2025); methodology and data review.
Oral presentations: PRI Innovation Forum, B-Link (May 2026); NSFC-RGC Conference on Frontiers of Industrial Big Data and Intelligent Systems, battery disassembly planning (Apr 2023).

DHC Software Co., Ltd. | Technical Assistant, Hohhot, China **Summers 2021 and 2022**

- Assisted with steel supports for LED displays and monitoring equipment, strength checks, stiffness improvements, site installation and delivery.

Patent: Co-inventor, automatic bolt and nut disassembly and sorting device. Chinese application 2023104488742; publication CN 116619301 A; filed April 24, 2023.
Honors and recognition: Outstanding Award, New Engineering Undergraduate Graduation Design Competition (Aug 2024); Provincial Completion Certificate, Chinese College Students Entrepreneurship Plan Competition (Jun 2024); College-Level Gold Award, Challenge Cup (Apr 2024); Third Prize, CTGU Internet+ Entrepreneurship Competition (Jul 2023).
Technical skills: Python, MATLAB, PyTorch, MuJoCo, NVIDIA DeepStream, TensorRT, SolidWorks, ANSYS Fluent/Workbench/STK.